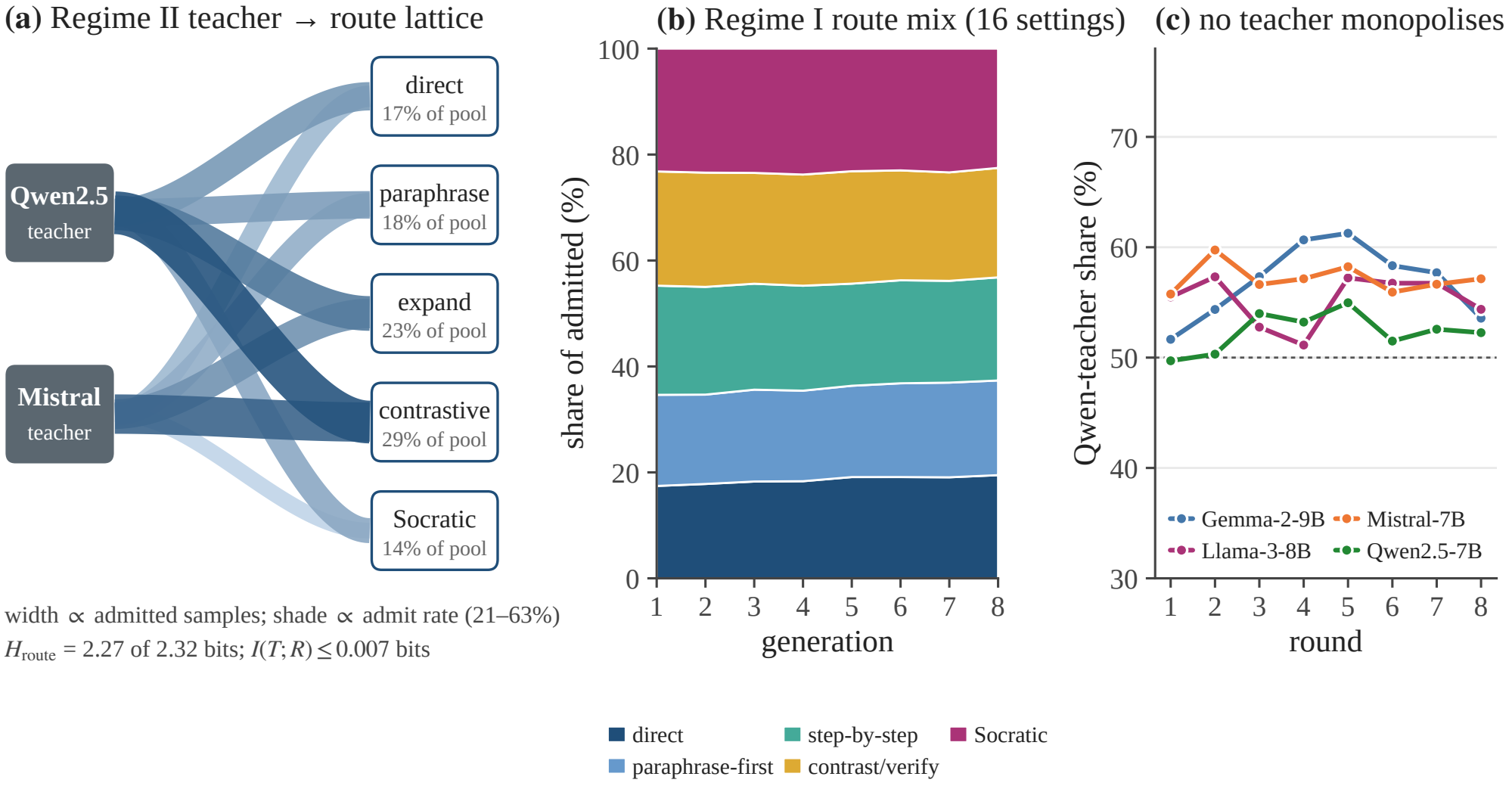
(a) Regime II teacher → route lattice
Qwen2.5
teacher
Mistral
teacher
direct
17% of pool
paraphrase
18% of pool
expand
23% of pool
contrastive
29% of pool
Socratic
14% of pool
width ∝ admitted samples; shade ∝ admit rate (21–63%)
H_{route} = 2.27 of 2.32 bits; $I(T;R) \leq 0.007$ bits
(b) Regime I route mix (16 settings)
share of admitted (%)
generation
direct
step-by-step
Socratic
paraphrase-first
contrast/verify
(c) no teacher monopolises
Qwen-teacher share (%)
round
Gemma-2-9B
Mistral-7B
Llama-3-8B
Qwen2.5-7B